Bookstore Trade-In - Lab 7 Fuzz Testing Report

Original failure reproduction, defensive fixes, property-based tests, and final verification

Project overview

The repository contains Lab 6 buyback and scanner-calibration functions and Lab 7 ISBN formatting, depreciation, and condition-multiplier functions. This report documents the original Lab 7 defects, input-validation changes, real terminal captures, and test results. The referenced Lab 7 handout was not present in the repository; the starter code and its README informed the tested contracts.

Fuzz testing and Hypothesis

Fuzz testing exercises software with generated inputs to expose edge cases not covered by a few examples. Hypothesis integrates with pytest, generates and shrinks values, and supports property-based contracts. The tests cover valid and malformed strings, empty and Unicode values, None and wrong types, negative and very large values, non-finite numbers, and boundary ratios. Invalid types deliberately raise TypeError; malformed values raise ValueError rather than succeeding or leaking incidental exceptions.

Bugs found and fixes

Area	Observed defect	Fix
ISBN-10	Empty input caused IndexError; invalid lengths, misplaced X, Unicode digits, and long strings were not consistently rejected. None caused AttributeError.	Type guard and strict ASCII ISBN-10 body format check; preserved the output shape. Checksum calculation remains out of scope.
Depreciation	Zero lifespan caused ZeroDivisionError; negative and non-finite values could produce invalid factors; None caused AttributeError.	Validate exactly one separator, numeric finite values, age >= 0, and lifespan > 0 before calculating the existing rounded factor.
Condition	Unknown/empty condition leaked KeyError; None caused AttributeError.	Type guard and explicit supported-code check, returning the existing multipliers or a clear ValueError.
CLI	Invalid input and terminal EOF could terminate the interactive run with an unhandled traceback.	Display validation/missing-input messages and exit cleanly.

Before-and-after results

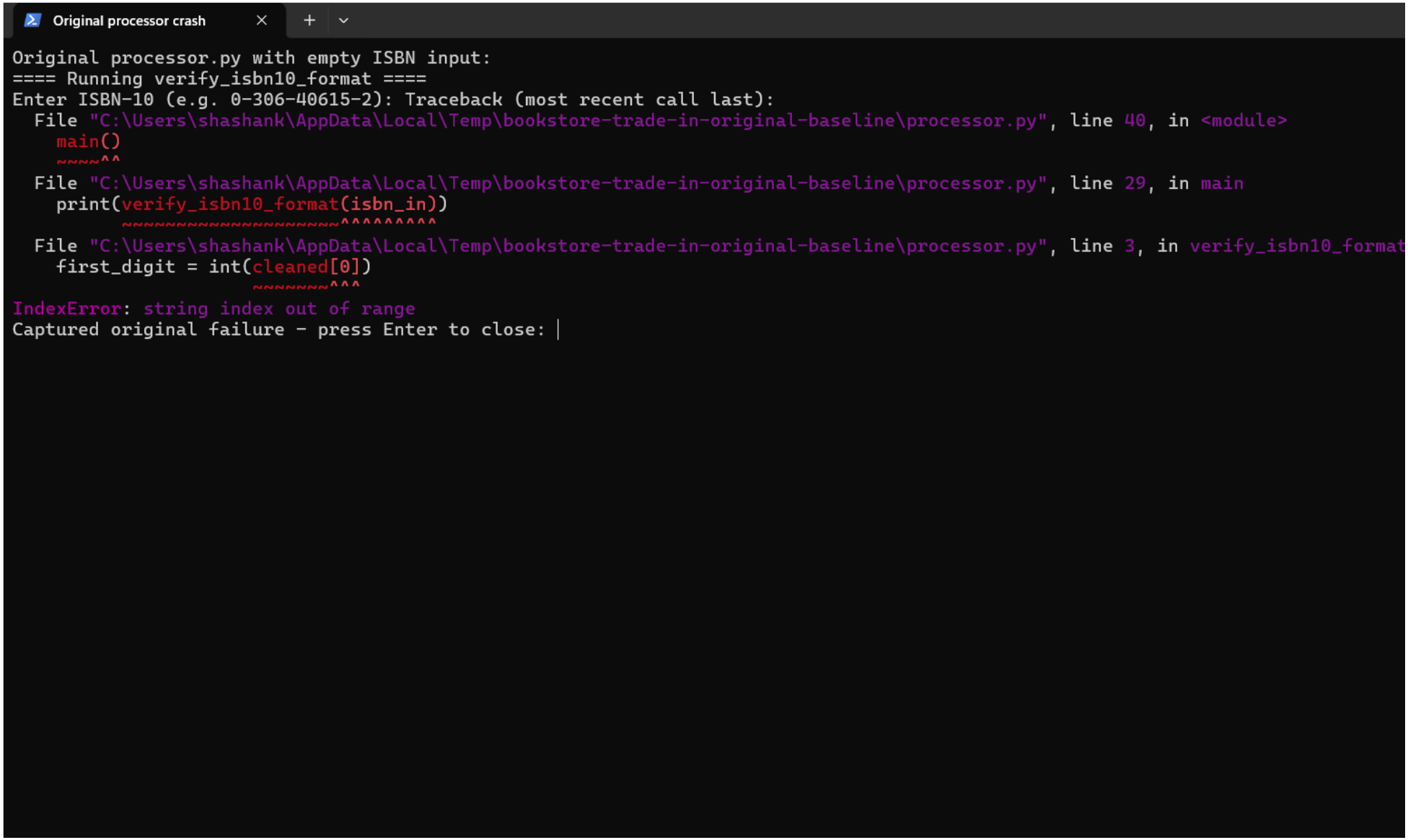
Run	Observed result
Original-code Hypothesis baseline	19 failed, 6 passed, 5 deselected (25 selected tests); reproduced against the original processor in an isolated temporary folder.
Fixed Lab 7	30 passed.
Lab 6 regression and coverage	12 passed; 100% line and branch coverage of bookstore_trade_in.py.

Run	Observed result
Final full repository suite	42 passed (rerun during this continuation).

Screenshots are genuine Windows Terminal captures made while running the original code and final tests. The baseline image shows the original version tested with the same 25 selected cases as the initial expanded run; five later-added deterministic boundary cases were deselected.

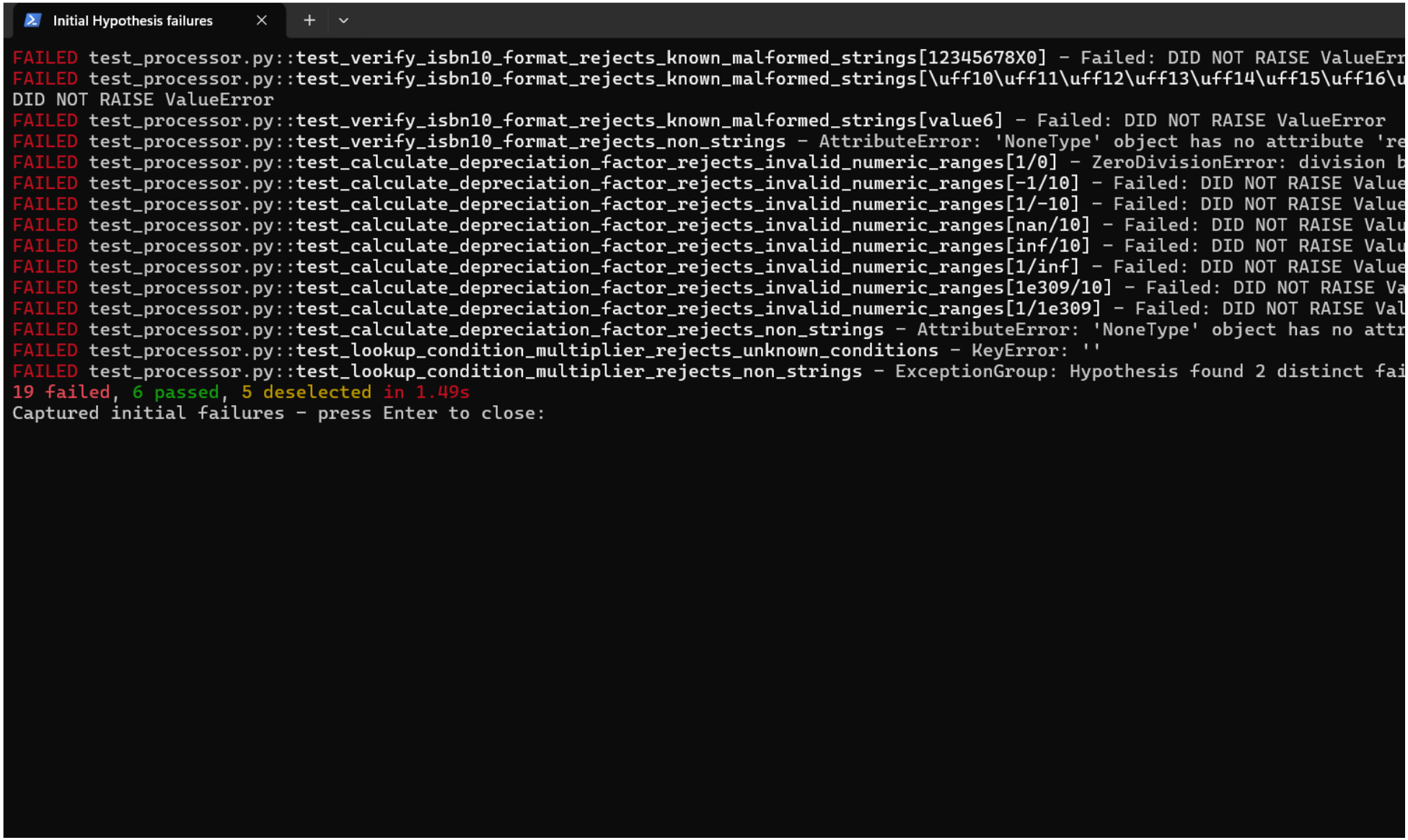
Original processor failure

Ran the original committed processor with an empty ISBN. The real traceback ends in IndexError while indexing the empty cleaned string.



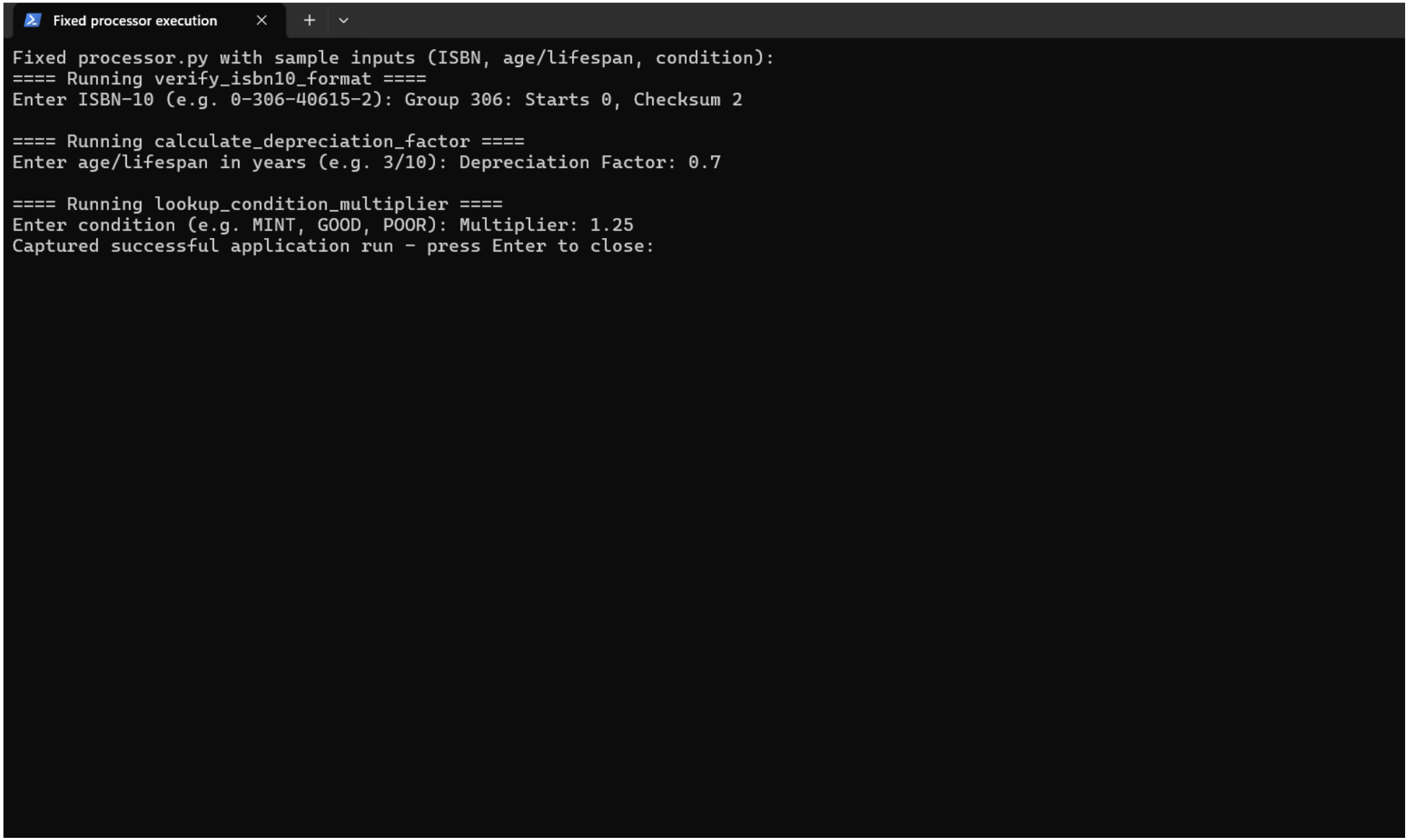
Initial Hypothesis failures

Reproduced the 25-case baseline against the original processor: 19 failed, 6 passed, and 5 later-added boundary cases deselected.



Fixed processor execution

Ran the updated interactive processor with ISBN 0-306-40615-2, age/lifespan 3/10, and condition MINT; it returned the expected summary and values.



Final full pytest run

Ran python -m pytest -v from the repository root; the terminal reports 42 passed.

